高分子化学学习笔记

李明翰

2026 年 3 月 13 日

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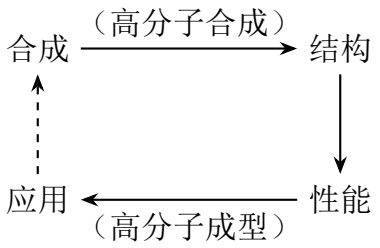
1 绪论

高分子化学 是高分子科学的基础，研究高分子化合物的分子设计合成及改性，研究聚合反应和高分子化学反应原理。担负着为高分子科学研究提供新生化合物为国民经济提供新材料及合成方法的任务。

高分子物理 高分子科学的理论基础。主要研究高分子机器聚集态结构、性能、表征以及结构与性能，结构与外立场的影响之间的相互关系，指导高分子化合物的分子设计和高聚物作为材料的合理使用，是沟通合成和应用的桥梁。

高分子工程 研究设计聚合反应工程、高分子成型工艺及相应原理的理论、方法的研究，是高分子科学与高分子工业之间的衔接点。



高分子化学是研究聚合物合成与化学反应的一门学科，涉及到两类反应：
聚合物的合成反应

$$小分子 \rightarrow 大分子$$

聚合物的化学反应

$$新的大分子 \leftarrow 大分子 \rightarrow 小分子$$

- 反应的机理
- 反应的影响因素
- 反应的动力学
- 反应的实施方法

1.1 高分子的基本概念

1.1.1 高分子的定义

高分子也称聚合物或者高聚物，是由很多相同的简单的结构单元通过共价键重复连接而成的，相对分子质量很大的大分子化合物。具体多大算大分子并无明确的界限

一般的高分子分子量在 10^4—10^7 之间。

- **单体**合成高分子的低分子量原料。

  尼龙-66 的单体

1 Introduction

Polymer chemistry is the foundation of polymer science, studying the molecular design, synthesis, and modification of polymer compounds, and researching polymerization reactions and the principles of polymer chemical reactions. It is responsible for providing new chemical compounds for polymer science research and supplying new materials and synthesis methods for the national economy.

Polymer Physics is the theoretical foundation of polymer science. It mainly studies the aggregated state structure, properties, and characterization of polymeric machines, as well as the relationships between structure and properties, and the influence of structure on external fields. It guides the molecular design of polymer compounds and the rational use of polymers as materials, serving as a bridge between synthesis and application.

Polymer engineering studies and designs polymer reaction engineering, polymer molding processes, and the corresponding theoretical and methodological research, serving as the connection between polymer science and the polymer industry.

$$\begin{tikzcd}
\text{合成} \arrow[r, "\text{（高分子合成）}"] & \text{结构} \arrow[d] \\
\text{应用} \arrow[u, dashed] & \text{性能} \arrow[l, "\text{（高分子成型）}"]
\end{tikzcd}$$

高分子化学是研究聚合物合成与化学反应的一门学科，涉及到两类反应：
聚合物的合成反应

$$\text{小分子} \rightarrow \text{大分子}$$

聚合物的化学反应

$$\text{新的大分子} \leftarrow \text{大分子} \rightarrow \text{小分子}$$

- 反应的机理
- Factors affecting the reaction
- 反应的动力学
- Implementation method of the reaction

1.1 The Basic Concepts of Polymers

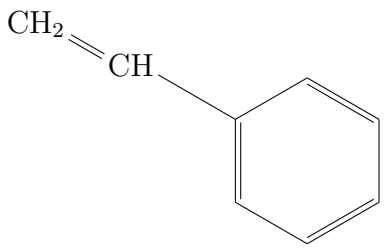
1.1.1 Definition of Polymers

Polymers, also known as macromolecules, are large molecular compounds with a very high relative molecular mass, formed by many identical simple structural units connected repeatedly by covalent bonds. There is no clear boundary for how large a molecule must be to be considered a macromolecule.

一般的高分子分子量在 10^6—10^7 之间。

- Low molecular weight raw materials for synthesizing polymers from monomers.

  Nylon-66 monomer



苯乙烯

$CH_2{=}CH{-}Cl$

氯乙烯

- **重复结构单元**

  大分子链上的化学组曾和结构均可重复出现的最小基本单元，可称为链结

- **结构单元**

  单体分子通过聚和以后进入大分子链的基本的单元。结构单元的元素组成可以与单体的元素组成相同，也可以不同。

- **单体单元** 分子通过聚合反应形成的元素组成与单体完全相同的结构单元

- **聚合度**

  聚合物分子量大小的一个指标

1.2 聚合物的分类和命名

1.2.1 聚合物的分类

- 依据材料的性能和用途分类

  橡胶

  塑料

  纤维

  涂料

  胶粘剂

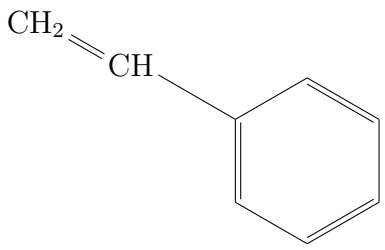
  功能高分子

- 依据高分子的主链结构分类

  碳链聚合物

  杂链聚合物

  元素有机聚合物



苯乙烯

CH_2 ~~CH~~ Cl 氯乙烯

- 重复结构单元

  The smallest basic unit on a macromolecular chain, whose chemical groups and structure can repeatedly

- 结构单元

  Monomer molecules become the basic units of macromolecular chains after polymerization. The elemental composition of the structural units can be the same as that of the monomers, or it can be different.

- 单体单元是指分子通过聚合反应形成的，其元素组成与单体完全相同的结构单元

- 聚合度

  A measure of the size of polymer molecules

1.2 Classification and Nomenclature of Polymers

1.2.1 Classification of Polymers

- Classified according to the performance and use of the materials

  Rubber, plastic, fiber, coating, adhesive, functional polymer

- Classification based on the main chain structure of polymers

  Carbon chain polymers, copolymers, elemental organic polymers

- 依据高分子受热后的形态变化分类

  热塑性高分子

  热固性高分子

1.2.2 聚合物的命名

- 依据材料的性能和用途分类
- 依据高分子的主链结构分类
- 依据高分子受热后的形态变化分类

1.2.3 聚合反应

定义 由小分子单体通过化学方法得到的高分子过程成为聚合反应

（1）按单体和聚合物的组成和结构变化分类

- 加聚反应
- 缩聚反应
- 加聚反应和缩聚反应的区别

（2）按聚合机理分类

- 连锁聚合
- 逐步聚合
- 连锁聚合与逐步聚合的比较

1.3 高分子化合物的基本特点

1.3.1 高分子的主链多由重复结构单元以共价键的形式相连接

1.3.2 分子量大

1.3.3 分子量有多分散性

1.3.4 分子形态呈多样性

1.3.5 具有显著的多层次结构

1.4 聚合热

热力学讨论范围：反应的可能性；反应的方向；反应平衡；反应热。

- Classification based on the morpholo gical changes of polymers after heati ng Thermoplastic polymers Thermos etting polymers

1.2.2 Naming of Polymers

- Classified according to the performance and use of the materials
- Classification based on the main chain structure of polymers
- Classification according to the morphological changes of polymers after heating

1.2.3 聚合反应

Definition: The process of obtaining polymers from small molecule monomers through chemical method

（1）Classified according to the composition and structural changes of monomers and polymers

- Addition reaction
- Condensation reaction
- The difference between addition polymerization and condensation polymerization

（2）按聚合机理分类

- 连锁聚合
- 逐步聚合
- Comparison of Chain Polymerization and Step-Growth Polymerization

1.3 Basic Characteristics of Macromolecular Compounds

1.3.1 The main chain of macromolecules is mostly connected by rep eating structural units in the form of covalent bonds 1.3.2 Large mol ecular weight 1.3.3 Molecular weight is polydisperse 1.3.4 Molecul ar morphology is diverse 1.3.5 Has significant multi-level structure

1.4 聚合热

Thermodynamics discussion scope: the feasibility of reactions; the direction of reactions; reaction equilib

1.4.1 热力学的一般概念

聚合热（焓） 因单体转变为聚合物是体系的无序度减小故 $\Delta S < 0$，而且各单体的聚合熵一般变化不大约为 $-105 \sim -125 jmol^{-1}K^{-1}$ 因此要使聚合反应进行必须

$$\Delta H < 0$$

$$|\Delta H| > |T\Delta S|$$

- 聚合反应必为放热反应
- 放热大小，差异很大。聚合热越大聚合倾向也越大。

烯类单体的聚合热 被破坏的是一个碳碳双键，重组的是两个碳碳单键则聚合热为 $-\Delta H = 90 kjmol^{-1}$ 单体结构还讲影响聚合热，通常

- 单体的键能越大，聚合热越小
- 聚合物的键能越大，聚合热越大

1) 取代基的位阻效应

   1,1-二取代的单体聚合热较小，聚合倾向较小

2) 共轭效应和超共轭效应

   共轭效应和超共轭效应使得单体的键能较大，聚合热较小，聚合倾向较小

3) 取代基的电负性

   取代基的电负性越大，单体的键能越小，聚合热越大，聚合倾向越大

4) 氢键以及溶剂化作用

   氢键以及溶剂化作用使得单体的键能较大，聚合热较小，聚合倾向较小

缩聚类单体的聚合热

1) 聚合前后总内能变化不大，实际聚合热低
2) 缩聚反应的单体极性高，分子聚集相对有规律
3) 当每个参加反应的的分子中的官能团数大于等于 2 时，才能能形成聚合物，为缩聚反应
4) 缩聚反应是通过官能团之间的多次缩合反应形成聚合物的过程

1.4.1 General Concepts of Thermodynamics

聚合热（焓）因单体转变为聚合物是体系的无序度减小故 $\Delta S < 0$，而且各单体的聚合熵一般变化不大约为 $-105 \sim -125 jmol^{-1}K^{-1}$ 因此要使聚合反应进行必须

$$\Delta H < 零$$

$$|\Delta H| > |T\Delta S|$$

- Polymerization reactions must be exothermic reactions
- The amount of heat released varies greatly. The greater the polymerization heat, the greater the tendenc

The polymerization heat of olefin monomers is such that what is broken is a carbon-carbon double bond, and what is reorganized are two carbon-carbon single bonds, so the polymerization heat is $-\Delta H = 90\ kjmol^{-1}$. The structure of the monomer also affects the polymerization heat, usually

- The larger the bond energy of the monomer, the smaller the polymerization heat.
- The greater the bond energy of the polymer, the greater the polymerization heat.

1) Steric effect of substituents

1,1-二取代的单体聚合热较小，聚合倾向较小

2) Conjugation effect and hyperconjugation effect

Conjugation effects and hyperconjugation effects make the bond energy of the monomer larger, the po

3) 取代基的电负性

The greater the electronegativity of the substituent, the smaller the bond energy of the monomer, the greater the polymerization heat, and the greater the polymerization tendency. 4) Hydrogen bonding and solvation effects

Hydrogen bonding and solvation cause the monomer to have higher bond energy, lower polymerizatio

Polymerization heat of condensation monomers

1) The total internal energy does not change much before and after polymerization, and the actual polymeriza

2) The monomers of condensation reactions have high polarity, and the molecular aggregation is relatively regular. 3) Polymers can only be formed when the number of functional groups in each participating molecule is greater than or equal to 2; this is a condensation reaction. 4) Condensation reaction is the process of forming polymers through multiple condensation reactions between functional groups.

环状单体的聚合热 开环聚合中，聚合前后键的类型不变，因此聚合热主要来自于环张力的释放。

环张力主要包括两类：

- 键角变形引起的角张力
- 氢或取代基间斥力造成的构象张力

对于环烷烃，不同环系的开环聚合能力可概括如下：

1) 三、四元环

角张力和聚合热都很大，容易发生开环聚合。

2) 五元环

键角约为 $108°$，角张力和 ΔH 都较小，ΔS 项对能否开环聚合起重要作用。

3) 六元环

常呈椅式或船式构象，键角变形趋于零，$\Delta H \approx 0$，ΔG 为正，因此一般无法聚合。

4) 八元以上环

环内氢或取代基处于拥挤状态，由于斥力形成跨环张力，即构象张力。

5) 十一元以上环

跨环张力基本消失，ΔH 和 ΔS 的贡献相近。

环烷烃的易开环程度可表示为：

$$3,4 > 8 > 7,5$$

Heat of polymerization of cyclic monomers In ring-opening polymerization, the types of bonds do not change before and after polymerization, so the heat of polymerization mainly comes from the release of ring strain.

Circumferential tension mainly includes two categories:

- Torsional stress caused by key angle deformation
- Conformational strain caused by repulsion between hydrogen or substituents

For cycloalkanes, the ring-opening polymerization ability of different ring systems can be summarized as

1) 三、四元环

The angle strain and polymerization heat are both large, making ring-opening polymerization easy to c

2) Five-membered ring

The bond angle is about 108°, the angular tension and ΔH are both small, and the ΔS term plays an in

3) Six-membered ring

Usually presents chair or boat conformations, with bond angle deformation approaching zero, $\Delta H \approx 0$, ΔG positive, therefore generally unable to polymerize.

4) Eight-membered or larger ring

The hydrogen or substituents within the ring are in a crowded state, and due to repulsion, transannular

5) Rings over eleven yuan

Cross-ring tension is basically gone, and the contributions of ΔH and ΔS are similar.

The ease of ring opening of cycloalkanes can be expressed as:

$$3, 4 > 8 > 7, 5$$

2 缩聚与逐步聚合

2.1 引言

缩合反应 有机官能团之间翻译除了形成著产物以外上有低分子物产生的反应，称为缩合反应。

- 参加缩合反应的反应物中，每个分子仅有一个参加反应的官能团
- 缩合反应只能形成低分子化合物
- 当每个参加反应的的分子中的官能团数大于等于 2 时，才能能形成聚合物，为缩聚反应
- 缩聚反应是通过官能团之间的多次缩合反应形成聚合物的过程

2.1.1 逐步聚合的基本概念

逐步聚合是高分子合成最基本的类型之一，逐步聚合的基本特征是官能团之间的反应。聚合度随时间逐步增长，而转化率在集合出去便可到达很高，从而表现出与连锁聚合不同的规律

2.1.2 逐步聚合的类型

（1）缩聚反应：官能团之间的缩合聚合反应，同时有小分子产生
（2）聚加成反应：形式上是加成反应，机理是逐步的。
（3）开环反应：部分为逐步反应
（4）氧化耦合：单体与氧气的缩合反应。
（5）Diel-Alder 反应

表 1: 逐步聚合与连锁聚合的比较

连锁聚合（Chain Polymerization）	**逐步聚合（Stepwise Polymerization）**
需活性中心：自由基、阳离子或阴离子，有链引发、增长、转移、终止等基元反应.	官能团间反应，无特定的活性中心，无链引发、增长、终止等基元反应.
单体一经引发，便迅速连锁增长，各步反应速率和活化能差别很大.	各反应逐步进行，每一步反应速率和活化能大致相同.
体系中只有单体和聚合物，无分子量递增的中间产物.	体系含单体和一系列分子量递增的中间产物.
转化率随着反应时间而增加，分子量变化不大.	分子量随着反应的进行缓慢增加，而转化率在短期内很高.

2 Condensation and Step-Growth Polymerization

2.1 Introduction

Condensation reaction refers to a reaction between organic functional groups in which, in addition to forming the main product, a small molecule is generated.

- In the reactants participating in the condensation reaction, each molecule has only one functional group
- Condensation reactions can only form low molecular weight compounds
- A polymer can only be formed in a condensation reaction when the number of functiona l groups in each reacting molecule is greater than or equal to 2.
- The polycondensation reaction is the process of forming polymers through multiple condensation reacti

2.1.1 基于步骤的聚合基本概念

Step-growth polymerization is one of the most fundamental types of polymer synthesis. T he basic characteristic of step-growth polymerization is the reaction between functional groups . The degree of polymerization gradually increases over time, and a high conversion rate can b e achieved when monomers are depleted, thus exhibiting different patterns from chain polymer ization.

2.1.2 Stepwise Polymerization Types

（1) Condensation reaction: A condensation polymerization reaction between functional groups, with the

（2）聚加成反应：形式上是加成反应，机理是逐步的。

（3) Open-loop response: part is a step response

（4）Oxidative coupling: the condensation reaction of monomers with oxygen.

（5）Diels-Alder 反应

Table 1: Comparison of Step-Growth and Chain-Growth Polymerization

连锁聚合（Chain Polymerization）	逐步聚合（Stepwise Polymerization）
需活性中心：自由基、阳离子或阴离子，有链引发、增长、转移、终止等基元反应.	官能团间反应，无特定的活性中心，无链引发、增长、终止等基元反应.
单体一经引发，便迅速连锁增长，各步反应速率和活化能差别很大.	各反应逐步进行，每一步反应速率和活化能大致相同.
体系中只有单体和聚合物，无分子量递增的中间产物.	体系含单体和一系列分子量递增的中间产物.
转化率随着反应时间而增加，分子量变化不大.	分子量随着反应的进行缓慢增加，而转化率在短期内很高.

2.2 缩聚反应

官能团之间进过多次缩合形成聚合物的反应

- 聚合物有特征结构官能团
- 有低分子副产物产生
- 缩聚物和单体分子量不成整数倍

2.2.1 官能度与平均官能度

官能团 单体分子中能参加反应并能表现出反应类型的原子团。— **官能度** 参加反应的官能团数。

平均官能度 两种或者两种以上的单体混合进行缩聚反应时，反应体系能参加反应的官能团数与单体总物质的量之比。对缩聚产物影响很大。

$$f = 1, \text{产物为低分子}$$

$$f = 2, \text{产物为线性聚合物}$$

$$f > 2, \text{产物为支链或者体型聚合物}$$

2.2.2 缩聚反应分类

a) 按产物的分子机构分类

- 线性缩聚反应
- 体型缩聚反应

b) 按参加反应的单体种类分类

- 自缩聚
- 杂缩聚

c) 按反应的热力学特征分类

- 平衡缩聚
- 不平衡缩聚

2.2 Condensation Reaction

官能团之间经过多次缩合形成聚合物的反应

- Polymers have characteristic structural functional groups
- There is the formation of low molecular by-products
- The molecular weight of the condensate and the monomer is not an integer multiple

2.2.1 Degree of Functionality and Average Degree of Functionality

Functional group: A group of atoms in a monomer molecule that can participate in reactio ns and exhibit types of reactions. — Functionality: The number of functional groups that partic ipate in reactions.

Average functionality When two or more types of monomers are mixed for a polyconden sation reaction, the ratio of the number of functional groups in the reaction system that can par ticipate in the reaction to the total amount of monomers. It has a great impact on the polyconde nsation product.

$$f = 1, \text{产物为低分子}$$

$$f = 2, \text{products are linear polymers}$$

$$f > 2, \text{products are branched or bulky polymers}$$

2.2.2 Classification of Condensation Reactions

a) Classification according to the molecular structure of the product

- Linear condensation reaction
- Polycondensation reaction of body size

b) Classified according to the types of monomers participating in the reaction

- Self-condensation
- co-polycondensation

c) Classify according to the thermodynamic characteristics of the reaction

- Balance condensation
- Unequal condensation

2.3 线性缩聚反应机理

2.3.1 线形缩聚与成环反应

缩聚反应过程中往往存在联众环化反应，单体单元内环化与分子内环化。

环化反应发生在同一单元体内，分子内环化发生在同一分子内的不同单元体之间。环化反应会降低聚合度，影响产物的性能。

2.3.2 线形缩聚机理

缩聚反应的重要特征之一是大分子链增长是逐步进行的。比如以 aAa 代表二元醇，bBb 他们形成聚酯的过程如下：

先两种分子单体相互反应生成二聚体

$$aAa + bBb \rightleftharpoons a\left[A-B\right]b + ab$$

二聚体与单体继续反应生成三聚体

$$a\left[A-B\right]b + bBb \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

或者二聚体相互反应生成四聚体

$$a\left[A-B\right]b + a\left[A-B\right]b \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

以此类推，逐步形成高分子链。此过程可表示为

$$n\text{聚体} + m\text{聚体} \rightleftharpoons (n+m)\text{聚体} + ab$$

研究缩聚反应的转化率没有意义，我们用反应程度 P 来表示缩聚反应的程度，定义为

$$P = \frac{n_0 - n}{n_0}$$

其中 $n_0 - n$ 表示反应过程中消耗的基团数，n_0 表示反应开始时基团的总数。聚合度与反应程度的关系为

$$\bar{X}_n = \frac{1}{1-P}$$

缩聚反应的另一特性是反应的可逆平衡性。

- k 小，聚酯类，小分子对聚合度影响较大。
- k 中，聚酰胺类，小分子对聚合度有所影响。
- k 大，可视为不可逆，酚醛体系。

2.3 Linear Condensation Reaction Mechanism

2.3.1 Linear Condensation and Cyclization Reactions

In the polycondensation reaction process, there are often concurrent cyclization reactions, including intra

Cyclization reactions occur within the same monomer unit, and intramolecular cyclization occurs between different monomer units within the same molecule. Cyclization reactions reduce the degree of polymerization and affect the properties of the product.

2.3.2 Linear Condensation Mechanism

One of the important characteristics of condensation reactions is that the growth of macrom olecular chains occurs step by step. For example, using aAa to represent a diol and bBb, the proc ess by which they form polyester is as follows:

Two types of molecular monomers first react with each other to form a dimer

$$aAa + bBb \rightleftharpoons a\left[A-B\right]b + ab$$

The dimer continues to react with the monomer to form a trimer

$$a\left[A-B\right]b + bBb \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

Or dimers react with each other to form tetramers

$$a\left[A-B\right]b + a\left[A-B\right]b \rightleftharpoons a\left[A-B\right]b\left[A-B\right]b + ab$$

By analogy, polymer chains are gradually formed. This process can be represented as

$$n\text{聚体} + m\text{聚体} \rightleftharpoons (n+m)\text{聚体} + ab$$

It is meaningless to study the conversion rate of the polycondensation reaction. We use the reaction extent P t

$$P = \frac{n_0 - n}{n_0}$$

其中 $n_0 - n$ 表示反应过程中消耗的基团数，n_0 表示反应开始时基团的总数。聚合度与反应程度的关系为

$$\bar{X}_n = \frac{1}{1-P}$$

Another characteristic of condensation reactions is the reversibility of the reaction equilibrium.

- k small, polyester, small molecules have a greater impact on the degree of polymerization.
- In k, polyamides, small molecules have some impact on the degree of polymerization.
- k large, can be considered irreversible, phenolic resin system.

在封闭体系中进行缩聚反应时，反应体系中小分子浓度会逐渐增加，导致反应达到平衡状态，限制了聚合度的进一步增长。因此，在实际操作中，常常通过不断移除生成的小分子（如水、醇等）来推动反应向生成高分子方向进行，从而获得更高的聚合度。

比如涤纶生产中，产率取决于真空度。

逐步特性是缩聚反应共有的，可逆的程度则取决于反应体系的热力学特征。

- 聚合物链逐步增长
- 反应呈现可逆平衡
- 往往有副反应

2.3.3 缩聚过程中的副反应

1) 基团消去反应

二元酸受热会脱羧，引起官能团数量变化。

2) 化学降解

缩聚反应是可逆的单体往往是聚合物的降解剂结果是分子量降低和分子量分布变宽。

3) 链交换反应

缩聚反应中形成的新官能团活性较大，容易与其他分子中的官能团发生反应，导致链交换反应，同种缩聚产物链交换结果是分子量分布变窄，不同缩聚产物链交换结果是产生嵌段共聚物。

2.4 线性缩聚反应动力学

2.4.1 官能团等活性概念

缩聚反应在形成大分子的过程中是逐步进行的若每一步都有不同的反应速率常数，缩聚反应的动力学分析将非常复杂。为简化分析，提出了官能团等活性概念，即假设所有参与反应的官能团具有相同的反应活性。这一假设使得缩聚反应的动力学分析变得可行，并且在许多实际情况下能够较好地描述反应过程。

Flory 提出了官能团等活性假设，认为官能团之间的碰撞次数和有效碰撞几率与高分子的扩散速率无关，体系粘度增大时虽然整个高分子运动速率减慢，但链段运动和链段的官能团活动并未受到限制。由于高分子的活动迟缓，扩散速率低反而是官能团之间碰撞时间变长，有利于提升反应速率。

官能团等活性假设 官能团的活性理论是近似的不是绝对的。

When performing a condensation reaction in a closed system, the concentration of small molecules in the reaction system gradually increases, causing the reaction to reach an equilibrium state and limiting further growth of the polymerization degree. Therefore, in practical operation, the reaction is often driven toward the formation of high polymers by continuously removing the generated small molecules (such as water, alcohol, etc.), thereby achieving a higher polymerization degree.

For example, in polyester production, the yield depends on the vacuum level.

Step-growth characteristics are common to condensation reactions, and the degree of reversibility depend

- Polymer chains grow step by step
- The reaction shows a reversible equilibrium
- There are often side effects

2.3.3 Side Reactions in Condensation Process

1) Group elimination reaction

Dicarboxylic acids will decarboxylate when heated, causing changes in the number of functional groups.

2) Chemical degradation

Condensation reactions are reversible. Monomers often act as degradants of polymers, resulting in a decrease

3) Chain exchange reaction

The new functional groups formed in the condensation reaction are highly reactive and can easily react with functional groups in other molecules, leading to chain exchange reactions. Chain exchange of the same condensation product results in a narrower molecular weight distribution, while chain exchange of different condensation products results in block copolymers.

2.4 Linear Condensation Reaction Kinetics

2.4.1 Functional Groups and Other Reactivity Concepts

In the process of forming macromolecules, polycondensation reactions proceed step by step. If each step has a different reaction rate constant, the kinetic analysis of polycondensation reactions will be very complicated. To simplify the analysis, the concept of functional group equivalence activity was proposed, that is, it is assumed that all functional groups involved in the reaction have the same reactivity. This assumption makes the kinetic analysis of polycondensation reactions feasible and can describe the reaction process well in many practical cases.

Flory 提出了官能团等活性假设，认为官能团之间的碰撞次数和有效碰撞几率与高分子的扩散速率无关，体系粘度增大时虽然整个高分子运动速率减慢，但链段运动和链段的官能团活动并未受到限制。由于高分子的活动迟缓，扩散速率低反而是官能团之间碰撞时间变长，有利于提升反应速率。

Functional group and other activity hypotheses The theory of functional group activity is approximate, n

2.4.2 不可逆的缩聚动力学

若将体系中的低分子副产物不断排出，则反应正向进行。可视为不可逆反应。以聚酯反应为例

$$-\frac{d[COOH]}{dt}=\frac{k_1k_3[COOH][OH][H^+]}{k_2k_{HA}}$$

外加酸催化体系

自催化缩聚反应的酯化速率太低。往往用酸催化剂。聚合过程中氢离子浓度可视为常数, 式子

$$-\frac{d[COOH]}{dt}=\frac{k_1k_3[COOH][OH][H^+]}{k_2k_{HA}}$$

其中所有的 k 可以与 $[H^+]$ 合并成一个常数 k'，得到

$$-\frac{dc}{dt}=k'c^2$$

变化可得

$$\frac{1}{c}-\frac{1}{c_0}=k't$$

引入反应程度 p，得到

$$\frac{1}{1-p}-1=k'c_0t$$

上式表明，外加酸催化缩聚反应的聚合度随聚合时间变化较快。通常外加酸催化缩聚反应的速率常数比自催化缩聚反应的速率常数大 1-2 个数量级。

自催化体系

当二元酸与二元醇中的两种基团数量相同，有无外加酸氢离子都来源于二元酸本身，分两种情况讨论：

1) 羧酸不电离

此时的聚酯化反应由二分子羧酸形成双分子络合，同时与 1 分子的羟基参与缩聚成为三级反应。速率方程为

$$-\frac{dc}{dt}=k''c^3$$

积分得到

$$\frac{1}{2}\left(\frac{1}{c^2}-\frac{1}{c_0^2}\right)=k''t$$

引入反应程度 p，得到

$$\frac{1}{(1-p)^2}-1=2k''c_0^2t$$

2.4.2 Irreversible Condensation Kinetics

If the low molecular by-products in the system are continuously removed, the reaction proceeds forward. It can be regarded as an irreversible reaction. Take polyester reaction as an example.

$$-\frac{d[COOH]}{dt} = \frac{k_1 k_3 [COOH][OH][H^+]}{k_2 k_{HA}}$$

Exogenous acid catalytic system

The esterification rate of the self-catalyzed polycondensation reaction is too low. Acid catalysts are often used. During the polymerization process, the concentration of hydrogen ions can be regarded as constant, formula

$$-\frac{d[COOH]}{dt} = \frac{k_1 k_3 [COOH][OH][H^+]}{k_2 k_{HA}}$$

All the k's here can be combined with [H^+] into a single constant k', resulting in

$$-\frac{dc}{dt} = k'c^2$$

变化可得

$$\frac{1}{c} - \frac{1}{c_0} = k't$$

Introducing the reaction extent p, we get

$$\frac{1}{1-p} - 1 = k'c_0 t$$

The above formula shows that the degree of polymerization of the acid-catalyzed polycondensation reaction changes rapidly with polymerization time. Generally, the rate constant of acid-catalyzed polycondensation reaction is 1-2 orders of magnitude larger than that of the self-catalyzed polycondensation reaction.

自催化体系

When the number of functional groups in a dibasic acid and a diol are the same, whether or not externally added acidic hydrogen ions are present, they all come from the dibasic acid itself, and two cases are discussed:

1) Carboxylic acids do not ionize

At this time, the polyesterification reaction forms a bimolecular complex from two molecules of carboxylic acid, while participating in condensation with one molecule of hydroxyl to become a tertiary reaction. The rate equation is

$$-\frac{dc}{dt} = k''c^3$$

Points earned

$$\frac{1}{2}\left(\frac{1}{c^2} - \frac{1}{c_0^2}\right) = k''t$$

Introducing the reaction extent p, we get

$$\frac{1}{(1-p)^2} - 1 = 2k''c_0^2 t$$

2.4.3 平衡缩聚动力学

聚酯化反应平衡常数非常小，小分子副产物若不及时排出，则逆反应不能忽视。令羧基起始浓度为 c_0，t 时刻时的浓度为 c，分别考虑水不排除和水完全排除两种情况：

$$-\mathrm{COOH} + -\mathrm{OH} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} -\mathrm{OCO}- + \mathrm{H_2O}$$

起始	c_0	c_0	0	0
t 时（水不排除）	c	c	$c_0 - c$	$c_0 - c$
t 时（水部分排除）	c	c	$c_0 - c$	n_w

封闭体系，水不排除是

$$-\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c)^2$$

带入 c=$c_0(1-p)$，得到

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}]$$

K 很大时第二项忽略。则积分得

$$\frac{1}{1-p} - 1 = k' c_0 t$$

水部分排除是

$$\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c) n_w$$

带入 c=$c_0(1-p)$，得到

$$\frac{dp}{dt} = k_1[c_0(1-p)^2 - \frac{p n_w}{K}]$$

同理 K 很大时第二项忽略。则积分得

$$\frac{1}{1-p} - 1 = k' c_0 t$$

2.5 线性缩聚物的聚合度

2.5.1 反应程度与平衡常数对聚合度的影响

封闭体系时，平衡时，令

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}] = 0$$

则

$$p = \frac{\sqrt{K}}{\sqrt{K}+1}$$

2.4.3 Balance Condensation Kinetics

The equilibrium constant for the polyesterification reaction is very small. If the small molecule by-products are not removed in time, the reverse reaction cannot be ignored. Let the initial concentration of the carboxyl group be c_0, and the concentration at time t be c, considering two cases: water is not removed and water is completely removed.

$$-\mathrm{COOH} + -\mathrm{OH} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} -\mathrm{OCO}- + \mathrm{H_2O}$$

起始	c_0	c_0	0	0
t 时（水不排除）	c	c	$c_0 - c$	$c_0 - c$
t 时（水部分排除）	c	c	$c_0 - c$	n_w

Closed system, water is not eliminated

$$-\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c)^2$$

Plug in c=$c_0(1-p)$, we get

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}]$$

When K is large, the second term can be ignored. Then the integral is

$$\frac{1}{1-p} - 1 = k' c_0 t$$

Partial water exclusion is

$$\frac{dc}{dt} = -k_1 c^2 + k_{-1}(c_0 - c) n_w$$

Substitute c=$c_0(1-p)$, we get

$$\frac{dp}{dt} = k_1[c_0(1-p)^2 - \frac{p n_w}{K}]$$

Similarly, when K is large, the second term can be ignored. Then the integral is

$$\frac{1}{1-p} - 1 = k' c_0 t$$

2.5 线性缩聚物的聚合度

2.5.1 Effect of Reaction Extent and Equilibrium Constant on Degree of Polymerization

When the system is closed, at equilibrium, let

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{p^2}{K}] = 0$$

则

$$p = \frac{\sqrt{K}}{\sqrt{K}+1}$$

$$\frac{1}{1-p} = \sqrt{K} + 1$$

部分排水体系，平衡时

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{pn_w}{c_0 K}] = 0$$

则

$$\bar{X_n} = \frac{1}{1-p} = \sqrt{\frac{c_0 K}{pn_w}} \propto \sqrt{\frac{K}{n_w}}$$

聚合物与平衡常数的平方根成正比，与含水量的平方根成反比。

- 平衡常熟很小的聚酯化反应，需高温减压充分脱出水分。
- 聚酰胺化反应，k=400 可以在稍低减压下允许较高水分残留
- K 很大时对聚合度要求不高，如可溶性酚醛树脂的预聚物可以在水介质中缩聚

2.5.2 基团数比对聚合度的影响

以上讨论都基于两基团等摩尔的条件。要满足这一条件，通常需要：

1) 单体高度纯化和精确计量；

2) 两基团同在一个单体分子上，如羟基酸、氨基酸；

3) 二元胺和二元酸先成盐。

许多实际操作中，往往在这些措施的基础上，再使某种二元单体略微过量，或另加少量单官能团物质封锁端基，以控制聚合度。为此，引入如下两个参数：

$$r = \frac{N_a}{N_b}$$

$$q = \frac{(N_b - N_a)/2}{N_a/2} = \frac{1-r}{r}$$

$$r = \frac{1}{q+1}$$

其中，N_a、N_b 分别为体系中官能团 a、b 的起始官能团数。

1．2-2 体系中基团数不等 设反应体系为

$$n\,aAa + m\,bBb \longrightarrow bB[AB]_n b$$

若设基团 a 的反应程度为 p，则：

$$\frac{1}{1-p} = \sqrt{K} + 1$$

部分排水系统，平衡时

$$\frac{dp}{dt} = k_1 c_0[(1-p)^2 - \frac{pn_w}{c_0 K}] = 0$$

则

$$\bar{X_n} = \frac{1}{1-p} = \sqrt{\frac{c_0 K}{pn_w}} \propto \sqrt{\frac{K}{n_w}}$$

The polymer is proportional to the square root of the equilibrium constant and inversely proportional to the sq

- The small polyesterification reaction in Changshu requires high temperature and reduced pressure to ful
- Polyamide formation reaction, k=400 can allow higher moisture content under slightly reduced pressur
- When K is large, the requirement for the degree of polymerization is not high; for example, the prepoly

2.5.2 The Influence of the Number of Functional Groups on the Degree of Polymerization

The above discussion is based on the condition of equimolar amounts of the two groups. To meet this co

1) Monomer high-purity purification and precise quantification;

2) Two functional groups are present on the same monomer molecule, such as hydroxy acids and amino acids

3) Binary amines and dicarboxylic acids first form salts.

In many practical operations, often on the basis of these measures, a certain binary mono mer is slightly in excess, or a small amount of mono-functional substance is additionally added to block the end groups, to control the degree of polymerization. For this purpose, the followin g two parameters are introduced:

$$r = \frac{N_a}{N_b}$$

$$q = \frac{(N_b - N_a)/2}{N_a/2} = \frac{1-r}{r}$$

$$r = \frac{1}{q+1}$$

Among them, N_a and N_b are the initial numbers of functional groups a and b in the system, respectively.

1. 在 2-2 体系中基团数不相等，设反应体系为

$$n\,aAa + m\,bBb \longrightarrow bB[AB]_n b$$

If the reaction extent of group a is p, then:

- a 的反应数为 $N_a p$，残留数为

$$N_a(1-p)$$

- b 的反应数同样为 $N_a p$，残留数为

$$N_b - N_a p$$

- 体系中残留端基总数为

$$N = N_a + N_b - 2N_a p$$

由于每一大分子链含有两个端基，因此大分子总数为端基数的一半，即

$$\frac{N_a + N_b - 2N_a p}{2}$$

体系中的结构单元总数为

$$\frac{N_a + N_b}{2}$$

因此，聚合度等于结构单元数除以大分子总数，即

$$\bar{X}_n = \frac{(N_a + N_b)/2}{(N_a + N_b - 2N_a p)/2} = \frac{1+r}{1+r-2rp}$$

考虑两种极限情况：

1) 当 $r = 1$ 或 $q = 0$ 时，

$$\bar{X}_n = \frac{1}{1-p}$$

2) 当 $p = 1$ 时，

$$\bar{X}_n = \frac{1+r}{1-r}$$

当 $r = 1$ 且 $p = 1$ 时，理论上聚合度趋于无穷大，即形成一个无限大的大分子；但实际上 p 不可能严格等于 1。

2. 2-2 体系中基团数相等时加入单官能团物质 若原本为两基团数相等的 2-2 体系，另外加入 N_b' 个单官能团物质 C_b，则有

$$r = \frac{N_a}{N_b + 2N_b'}$$

$$q = \frac{2N_b'}{N_a}$$

- a 的反应数为 $N_a p$，残留数为

$$N_a(1-p)$$

- b The number of reactions is also $N_a p$, and the number of residues is

$$N_b - N_a p$$

- The total number of residual end groups in the system is

$$N = N_a + N_b - 2N_a p$$

Since each macromolecular chain contains two end groups, the total number of macromolecules is half th

$$\frac{N_a + N_b - 2N_a p}{2}$$

The total number of structural units in the system is

$$\frac{N_a + N_b}{2}$$

Therefore, the degree of aggregation equals the number of structural units divided by the total number of

$$\bar{X}_n = \frac{(N_a + N_b)/2}{(N_a + N_b - 2N_a p)/2} = \frac{1+r}{1+r-2rp}$$

Consider two extreme cases:

1) 当 $r =$ 为 1 或 $q =$ 为 0 时,

$$\bar{X}_n = \frac{1}{1-p}$$

2) 当 $p =$ 为 1 时,

$$\bar{X}_n = \frac{1+r}{1-r}$$

When $r =$ is 1 and $p =$ is 1, theoretically the degree of polymerization tends to infinity, f orming an infinitely large macromolecule; however, in reality, p cannot be strictly equal to 1.

2. 2-2 体系中基团数相等时加入单官能团物质 若原本为两基团数相等的 2-2 体系，另外加入 N_b' 个单官能团物质 C_b，则有

$$r = \frac{N_a}{N_b + 2N_b'}$$

$$q = \frac{2N_b'}{N_a}$$

式中的“2”表示：一个单官能团物质 C_b 所造成的 b 官能团过量，与一个含两个 b 官能团的 bBb 分子所造成的过量是等效的。因为过量的 bBb 中，只有一个基团 b 参与封端作用，另一个基团并不起继续增长链段的作用，因此可视作类似于单官能团物质 C_b。

此时同样有

$$\bar{X}_n = \frac{1+r}{1+r-2rp} = \frac{q+2}{q+2(1-p)}$$

由此可见，轻微的计量失衡或少量单官能团封端剂的加入，都会显著影响缩聚反应的聚合度，因此在缩聚体系中，原料纯度、官能团计量比及封端剂用量都必须严格控制。

2.6 线性缩聚物的分子量分布

2.7 体型缩聚和凝胶化作用

2.8 缩聚和逐步缩聚的实施方法

2.9 重要的缩聚物和其他逐步缩聚物

In the formula, “ 2 ” means: the excess of b functional groups caused by a monofunctional substance C_b is equivalent to the excess caused by a molecule bBb containing two b functional groups. Because in the excess bBb, only one group b participates in the end-capping, while the other group does not play a role in further chain growth, it can therefore be regarded as similar to a monofunctional substance C_b.

At this time there is also

$$\bar{X}_n = \frac{1+r}{1+r-2rp} = \frac{q+2}{q+2(1-p)}$$

It can be seen that slight measurement imbalances or the addition of a small amount of monofunctional end-capping agent can significantly affect the degree of polymerization of the polycondensation reaction. Therefore, in a polycondensation system, the purity of the raw materials, the stoichiometric ratio of functional groups, and the amount of end-capping agent must be strictly controlled.

2.6 Molecular Weight Distribution of Linear Condensates

2.7 Body Shrinking and Gelation Effects

2.8 Implementation methods of condensation and step-growth polymerization

2.9 重要的缩聚物和其他逐步缩聚物